

References

- [1] Shannon, C.E.
A Mathematical Theory of Communication.
Bell System Technical Journal, **27**(3), 379–423, 1948.
- [2] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000.
- [3] Cover, T.M. and Thomas, J.A.
Elements of Information Theory (2nd ed.).
Wiley Series in Telecommunications and Signal Processing. Wiley-Interscience, 2006.
- [4] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. *arXiv preprint arXiv:2001.08361*, 2020.
- [5] Giraldo, L.G.S., Rao, M., and Principe, J.C.
Measures of entropy from data using infinitely divisible kernels.
IEEE Transactions on Information Theory, 2014.
- [6] Ohad Shamir, Sivan Sabato, and Naftali Tishby. Learning and generalization with the information bottleneck. *Theoretical Computer Science*, 411(29):2696–2711, 2010.
- [7] DeepSeek AI. *DeepSeek-RL: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning.* *arXiv preprint arXiv:2501.12948*, 2025.
- [8] Tishby, N. and Zaslavsky, N.
Deep Learning and the Information Bottleneck Principle.
arXiv preprint arXiv:1503.02406.
- [9] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. *arXiv preprint arXiv:2203.02155*, 2022.
- [10] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? *arXiv preprint arXiv:2504.13837*, 2025.
- [11] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I.
Language Models are Unsupervised Multitask Learners.
OpenAI Technical Report, 2019.
- [12] Skean, O., Osorio, J.K.H., Brockmeier, A.J., and Giraldo, L.G.S.
DiME: Maximizing Mutual Information by a Difference of Matrix-Based Entropies.
arXiv preprint arXiv:2302.13949, 2023.
- [13] Grégoire Delétang, Anian Ruoss, Paul-Ambroise Duquenne, Elliot Catt, Tim Genewein, Christopher Mattern, Jordi Grau-Moya, Li Kevin Wenliang, Matthew Aitchison, Laurent Orseau, Marcus Hutter, and Joel Veness. Language modeling is compression. *arXiv preprint arXiv:2309.10668*, 2024.
- [14] Oscar Skean and Md Rifat Arefin and Dan Zhao and Niket Patel and Jalal Naghiyev and Yann LeCun and Ravid Shwartz-Ziv
Layer by Layer: Uncovering Hidden Representations in Language Models
arXiv preprint arXiv:2502.02013.
- [15] Ravid Shwartz-Ziv and Naftali Tishby
Opening the Black Box of Deep Neural Networks via Information
arXiv preprint arXiv:1703.00810

- [16] Zeyuan Allen-Zhu and Yuanzhi Li
Physics of Language Models: Part 3.1, Knowledge Storage and Extraction
arXiv preprint arXiv:2309.14316
- [17] Yuzhen Huang and Jinghan Zhang and Zifei Shan and Junxian He
Compression Represents Intelligence Linearly
arXiv preprint arXiv: 2404.09937
- [18] Oscar C González, Yury Sokolov, Giri P Krishnan, Jean Erik Delanois, Maxim Bazhenov
Can sleep protect memorise from catastrophic forgetting? eLife 9:e51005
- [19] Tadros, T., Krishnan, G.P., Ramyaa, R., Bazhenov, M
Sleep-like unsupervised replay reduces catastrophic forgetting in artificial neural networks
Nature Communications, **13**, 7742 (2022)
- [20] Yu, F., Arora, H.S., and Johnson, M.
Iterative Graph Alignment.
arXiv preprint arXiv:2408.16667 (2024).
- [21] Fangyuan Yu. Scaling LLM pre-training with vocabulary curriculum. *arXiv preprint*
arXiv:2502.17910, 2025.
- [22] Berglund, L., Tong, M., Kaufmann, M., Balesni, M., Stickland, A.C., Korbak, T., and Evans, O.
The Reversal Curse: LLMs Trained on "A is B" Fail to Learn "B is A".
arXiv preprint arXiv:2309.12288 (2024).
- [23] Keller Jordan, Jeremy Bernstein, Brendan Rappazzo, @fernbear.bsky.social, Vlado Boza, Jiacheng You, Franz Cesista, Braden Koszarsky, and @Grad62304977.
modded-nanogpt: Speedrunning the NanoGPT baseline, 2024.
<https://github.com/KellerJordan/modded-nanogpt>.

A Appendix

Theorem 3 (Minimum Probability Entropy Bound). *Let X be a discrete random variable with sample space Ω , where $|\Omega| = n$. Suppose there exists a constant $\alpha \in (0, \frac{1}{n}]$ such that $P(X = x) \geq \alpha$ for all $x \in \Omega$. Then the entropy of X is bounded below by:*

$$H(X) \geq -(1 - \alpha(n - 1)) \log_2(1 - \alpha(n - 1)) - (n - 1)\alpha \log_2(\alpha).$$

Furthermore, for sufficiently large n and small α such that $\beta = \alpha n \ll 1$, this bound approximates to:

$$H(X) \geq \beta \log_2(n)$$

Proof. Under the constraint that $P(X = x) \geq \alpha$ for all $x \in \Omega$, the entropy

$$H(X) = - \sum_{x \in \Omega} P(x) \log_2 P(x)$$

is minimized when the distribution is as imbalanced as possible: one outcome has the highest allowed probability, and all others are assigned the minimum α . The worst-case distribution is:

$$\begin{aligned} P(x_1) &= 1 - \alpha(n - 1), \\ P(x_i) &= \alpha \quad \text{for } i = 2, \dots, n. \end{aligned}$$

The resulting entropy is:

$$H(X) = -(1 - \alpha(n - 1)) \log_2(1 - \alpha(n - 1)) - (n - 1)\alpha \log_2(\alpha).$$

For small α and large n such that $\alpha n \ll 1$, we approximate have:

$$1 - \alpha(n - 1) \approx 1 - \alpha n.$$

Substituting this into the expression:

$$\begin{aligned} H(X) &\approx -(1 - \alpha n) \log_2(1 - \alpha n) - \alpha n \log_2(\alpha) \\ &= -(1 - \alpha n) \log_2(1 - \alpha n) - \alpha n \log_2(\alpha n) + \alpha n \log_2(n) \\ &= h(\alpha n) + \alpha n \log_2(n) \end{aligned}$$

where $h(p) = -p \log_2 p - (1 - p) \log_2(1 - p)$ is the binary entropy function.

Since $h(\alpha n) \geq 0$ and $\log_2(1/\alpha) \geq \log_2(n)$ for $\alpha \leq 1/n$, we get:

$$H(X) \geq \alpha n \log_2(n) = \beta \log_2(n)$$

which completes the proof. $\square$

Corollary 1 (Entropy Lower Bound for Finite Discrete Random Variables). *Let X be a discrete random variable with finite support Ω , where $|\Omega| = n$, and assume that $P(X = x) > 0$ for all $x \in \Omega$. Then there exists a constant $\beta \in (0, 1]$ such that:*

$$H(X) \geq \beta \cdot \log_2 |\Omega|.$$

Proof. Let $\varepsilon := \min_{x \in \Omega} P(X = x)$, which exists and is strictly positive since X is discrete with finite support. By the Minimum Probability Entropy Bound (Theorem 3), we have:

$$H(X) \geq -(1 - \varepsilon(n - 1)) \log_2(1 - \varepsilon(n - 1)) - (n - 1)\varepsilon \log_2(\varepsilon).$$

For small ε and large n , this approximates to:

$$H(X) \geq \varepsilon n \log_2\left(\frac{1}{\varepsilon}\right),$$

which is linear in $n = |\Omega|$. Setting $\beta := \varepsilon n$, the result follows:

$$H(X) \geq \beta \cdot \log_2 |\Omega|.$$

$\square$

Broader Impact and Limitations

Limitations. While theoretically grounded, our proposed GAPT algorithm remains relatively simple. On large-scale pretraining tasks, it achieves less than a 5% improvement over the baseline, falling short of fully realizing the theoretical benefits of compact representations for generalization as suggested by Theorem 1. Furthermore, scaling experiments are missing, which requires more computation resources but may realize more potential in this new dimension for improving generalization. Additionally, entropy minimization is only one approach to compression; other mechanisms—such as sparse activation—may offer stronger biological plausibility and efficiency. A further limitation is observed in the arithmetic generalization task, where GAPT exhibits instability in out-of-distribution performance across different training runs. There could be some data selection side piece missing for stabilizing generalization-oriented training runs.

Broader Impact. Enhancing generalization in AI systems holds the potential to enable models that can extrapolate, reason, and discover novel knowledge—key milestones toward artificial general intelligence. However, systems that generalize without explainability may pose greater risks than those that merely memorize. We caution that aggressive compression of internal representations, while beneficial for efficiency and abstraction, may also increase susceptibility to unintended behaviors such as spurious generalization or adversarial misuse.